ANALYTICS CASE STUDY

Telecom Customer Churn Analysis

Diagnosing $1.27M Revenue Leakage & Building a Retention Strategy

Dataset: 7,043 customers | **Tools:** Python (Pandas) · Power BI

26.54%	$1.27M	3
Churn Rate	Revenue at Risk	Key Segments

Harsh Anvekar · Data Science & BI · 2025

Churn is not random — it is concentrated in three identifiable segments

26.54%

Overall Churn Rate

1,869 of 7,043 customers

$11.8M

Current Annual Revenue

At present churn levels

$1.27M

Recoverable Revenue

If churn reduced by 30%

3

High-Risk Segments

Drive 80%+ of churn volume

M2M

Month-to-Month Contracts

42.7% churn rate — 7x higher than 2-year contracts. Contribute 55% of all churn volume.

FBR

Fiber Internet Customers

30-35% churn despite highest monthly charges. Represent 40% of total revenue leakage.

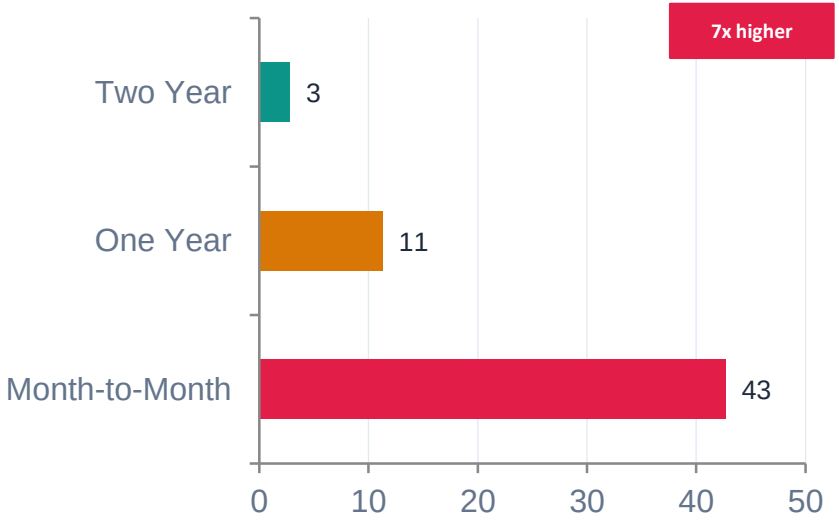
ADD

Customers with No Add-Ons

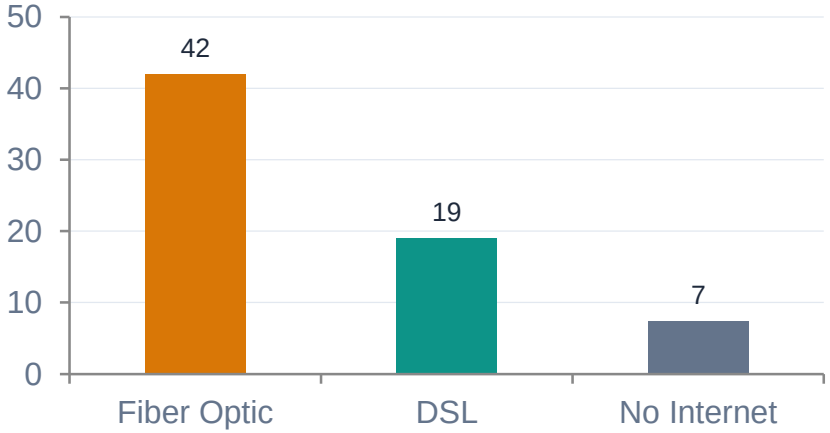
Churn 2-4x higher than customers with add-ons. Low service stickiness drives easy exits.

Two segments account for the majority of churn events and revenue loss

Contract Type vs Churn Rate



Internet Service Type vs Churn



Fiber customers = 40% of total revenue leakage despite being the highest-paying segment

Add-on adoption and payment method are the two strongest behavioural churn signals

Add-On Adoption Reduces Churn 2-4x

~25%

No Add-Ons

High exit risk

vs

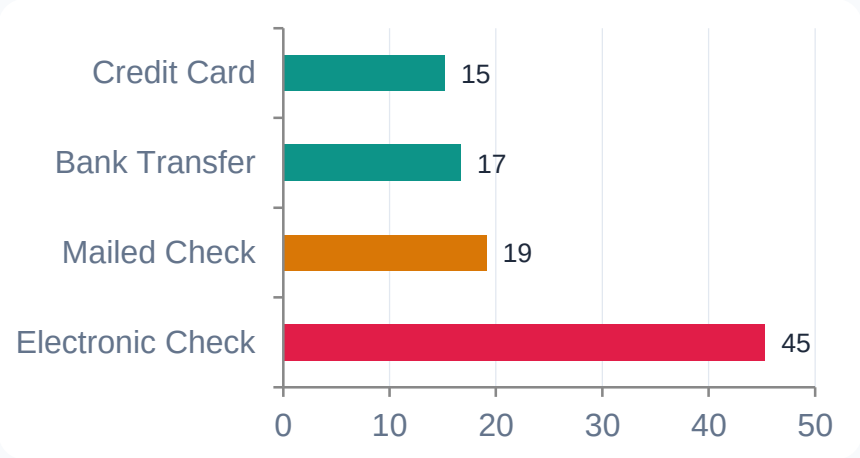
10-15%

With Add-Ons

Anchored customers

Security, backup, and tech support add-ons create multi-touchpoint lock-in. Converting zero-add-on users to even one bundle reduces exit probability by 2x.

Payment Method as a Churn Proxy



Electronic check users churn at 3x the rate of card/auto-pay users. Payment friction is a proxy for billing dissatisfaction and default risk.

A 30% reduction in churn unlocks ~$1.27M in annual revenue — a ~10% uplift

$11.8M

Current Revenue

At 26.54% churn

$1.27M

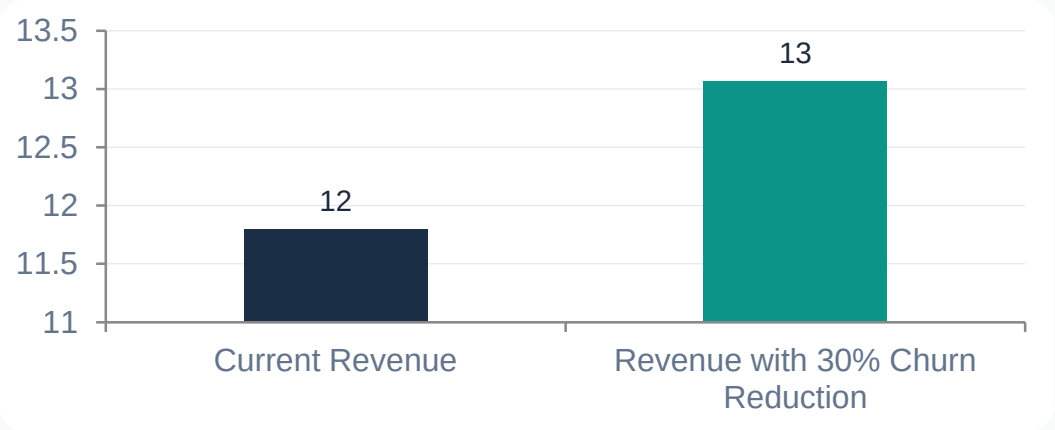
Revenue at Risk

From churned customers

$13.07M

Revenue Potential

At 30% churn reduction



Revenue Uplift Breakdown

Recover M2M churners (55%)	+$0.70M
Fiber segment (40%)	+$0.51M
Add-on conversion uplift	+$0.06M
Total recoverable	**$1.27M**

Three high-ROI interventions to recover $1.27M in annual revenue

01

Introduce Soft Lock-In Contracts

Target: Month-to-Month Customers (55% of churn)

- Offer 6-12 month contracts with 5-8% loyalty discounts
- Bundle perks (priority support, data rollover) to reduce friction
- Flag at-risk M2M customers in CRM for proactive outreach

Reduce M2M churn from 42.7% to ~30%

02

Proactive Fiber Experience Program

Target: Fiber Optic Customers (40% of revenue leakage)

- Implement automatic outage detection with proactive alerts
- Assign dedicated support tiers for fiber customers
- Reduce issues via predictive network maintenance

Reduce Fiber churn from ~42% to ~28%

03

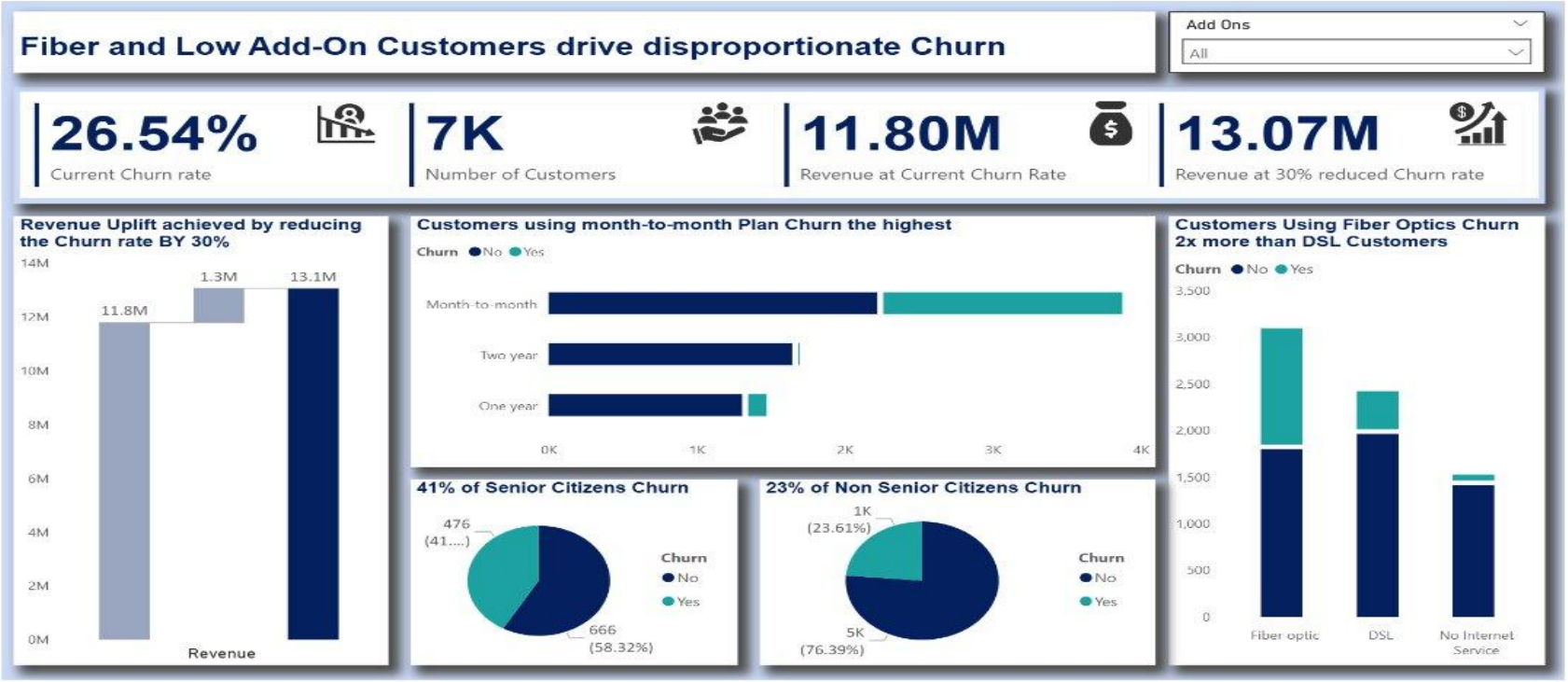
Add-On Bundle Campaign

Target: Zero Add-On Customers (2-4x churn risk)

- Offer discounted Security + Backup + TechSupport starter pack
- Target new signups and M2M customers in first 90 days
- Track add-on adoption as a leading retention KPI

Convert 20% of zero-add-on base to bundled plans

Interactive Power BI dashboard built to support executive decision-making



Dashboard includes Add-Ons slicer, KPI cards, revenue waterfall, contract-type breakdowns, and demographic churn splits